2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON) | 979-8-3315-3435-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/BECITHCON64160.2024.10962674

2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)
28-29 November 2024, Southeast University, Dhaka, Bangladesh

Machine Learning Based Earlier Pancreatic Cancer Detection Using Hybrid Feature Selection

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***Abstract*— Pancreatic Ductal Adenocarcinoma Cancer (PDAC), a dangerous cancer in the world, has a very poor percentage of surviving more than 5 years. Usual treatments like chemotherapy, surgery, and radiation have not been shown to significantly improve survival. If we can simplify the diagnosis of diseases through Machine Learning (ML), then the condition of Pancreatic Ductal Adenocarcinoma Cancer in humans can be known much earlier. In this research, a machine learning-based model is proposed to detect PDCA at an early stage. Different preprocessing approaches are used to increase the model's efficiency. Chi-squared and SelectKBest are used to get the best features. Seven different ML classifiers, such as Extreme Gradient Boosting (XGBoost), LightGBM (LGBM), K-Nearest Neighbor (KNN), Decision Tree (DT), Random Forest (RF), and Extra Tree Classifier (ET), are utilized to find the best one for the prediction. Finally, it is observed that LGBM shows the best performance among others, with an accuracy of 97%.**

Keywords*— *Ductal Adenocarcinoma Cancer; Machine Learning; Preprocessing; Chi-squared and SelectKBest; LGBM

I. INTRODUCTION

Cancer is a serious disease that can affect both adults and kids. The World Health Organization (WHO) says it happens when normal cells turn into tumor cells, which then become malignant tumors. This change is caused by a mix of genetic factors and things like chemicals and other environmental factors. The WHO also found that in countries with less money, there's a higher chance of new cancer cases – about 81% more than in wealthier countries. In 2018, around 18.1 million people got diagnosed with cancer, and sadly, 9.6 million lost their lives because of it [1]. Pancreatic cancer happens when unusual cells start growing in the pancreas. The pancreas is about 6 to 10 inches long and flat. This type of cancer is more common in older people, with 90% of new cases occurring in individuals above 55 years old [2, 3]. Males face a greater risk of being affected compared to females [3]. Presently, no efficient biomarkers are available for detecting PDAC (Pancreatic Ductal Adenocarcinoma) earlier. The sole biomarker in clinical use, serum CA19-9, lacks the requisite specificity and sensitivity for screening purposes [4].

Furthermore, the survival rate of individuals affected is low. As reported in the GLOBOCAN 2018 data, the incidence of pancreatic cancer amounted to 458,918 new cases, with a corresponding 432,242 fatalities. This data underscores that pancreatic cancer contributes to 4.5% of the overall mortality associated with cancer-related deaths [5, 6].

Pancreatic cancer causes are not fully understood; indeed, many factors have been identified that raise the risk of developing the disease including smoking, drinking too much alcohol, having a family history of the disease, genetic factors, the way you eat, having diabetes, long-term pancreas inflammation, and the presence of Helicobacter pylori [3, 7].

Therefore, finding pancreatic cancer early is hard. Because it often gets diagnosed late, the main way to prevent it is by living a healthy lifestyle. This really helps lower the chances of getting the disease to find pancreatic cancer; doctors usually start with a CT scan, followed by a test called endoscopic retrograde cholangiopancreatography. More accurate methods like endoscopic ultrasound and MRI are also used. If pancreatic cancer is confirmed, additional tests are needed [8]. Additionally, doctors use biomarkers like liquid biopsy for detection. They also used a blood test called CA19-9 to check the disease [3].

Different ML models are used to diagnose different diseases [9, 10, and 11]. Pancreatic cancer can be detected at an early stage using machine learning [12, 13, 14, and 15]. In this work, we proposed a model to detect pancreatic cancer in a more accurate way. The main contributions to this work are listed as follows:

- Proposed a model to predict pancreatic cancer using machine learning.
- Different preprocessing approaches, such as label encoding, null value handling, and optimal feature selection, are used to increase the model's performance.
- To obtain the best features, SelectKBest and Chi-squared are utilized.
- Different metrics, such as Accuracy, Precession, Recall and F-1 score, are used to evaluate different ML models.
- Finally, we obtained LGBM with an accuracy of 97% that outperformed other related works to predict pancreatic cancer.

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II. RELATED WORKS

Classification is like sorting things into groups based on labels or categories. It is a type of supervised learning, where the goal is to train the data using these labels to achieve a certain level of accuracy. Aiming to achieve a specific level of accuracy to predict pancreatic cancer. Sadewo W. et al. [12] used Twin SVM based on the kernel, resulting in an accuracy of 98%. The utilization of the Twin Support Vector Machine (TWSVM) method based on the Radial Basis Function (RBF) kernel proves effective in early detection of pancreatic cancer through blood tests. This particular kernel demonstrates superior performance, achieving an accuracy of 98%, sensitivity of 97%, and 100% specificity. The associated running time for this method is recorded at 1.3408 seconds.

Qiu Y. et al. [5] applied various machine-learning methods, including the decision tree (DT), KNN, and SVM. The outcomes of the implementation revealed that the SVM method exhibited the most effective performance, achieving 70% sensitivity and 70% specificity.

Debernardi S. et al. [13] used a group of urinary biomarkers to detect pancreatic cancer at an earlier stage. They wanted to check how accurate a new panel is replacing REG1A with REG1B and applying the PancRISK score algorithm they looked at more urine samples collected in the past. They also checked how well this new panel works with CA19-9, and how the biomarkers perform in common urinary tract cancers, considering their daily changes and stability. The study also demonstrated that the biomarkers do not significantly vary.

Yokoyama S. et al. [14] applied Machine learning models using SVM and NNET can tell apart pancreatic cancer patients after surgery, predicting different survival times. But we need more research to check if these models can help decide when to schedule follow-up appointments after surgery.

Setiawan Q. et al. [15] applied DT and Logistic Regression to predict pancreatic cancer. These two methods were compared to find out which one performs better in terms of accuracy. The findings indicated that the DT and Logistic Regression achieved their highest accuracy at 100% and 92.68%, respectively. Consequently, the Decision Tree is considered a better method, as it showed higher accuracy in classifying pancreatic cancer.

After looking at what other researchers have done, we noticed that different methods give different results for predicting pancreatic cancer. Since pancreatic cancer is a serious concern, we want to predict it as accurately as possible. Therefore, we are motivated to work on pancreatic cancer research to get better results. In our study, we have actually achieved better results compared to what others have done.

III. MATERIALS AND METHODS

In this section, the experiment's proposed strategy and the techniques utilized for the identification of pancreatic cancer are detailed. "Fig. 1" clearly shows architecture of the proposed model. First stage, we collect the data from Kaggle and the next stage we preprocess them by level encoding, null value handling and selecting optimal feature. Then we take 80% data for training and 20% data for testing. By using different types of classifiers, we are trying to detect pancreatic cancer early. We want to find out, the best classification algorithms from machine learning to help doctors identify people with pancreatic cancer accurately. In Our research, we use Extra Tree Classifier, XGBoost, RN, NB, DT, KNN & LGBM classifiers.

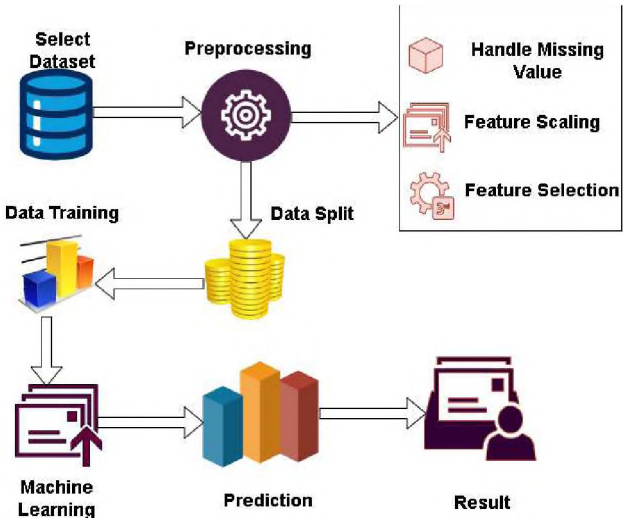


Fig. 1: The structure of the proposed model

A. Dataset

In this research, to detect Pancreatic Ductal Adenocarcinoma Cancer (PDAC) 'Urinary biomarkers for pancreatic cancer' dataset is used. In dataset we have 590 samples. In the data set, there are 14 features, which includes sample_id, patient_cohort, sample_origin, age, sex, diagnosis, stage, benign_sample_diagnosis, plasma_CA19_9, creatinine, LYVE1, REG1B, TFF1, REG1A. Before applying a classifier, we preprocess the data so that we can get best accuracy.

B. Level Encoding

Label encoding is a process of converting categorical data into numerical form. In machine learning, many algorithms require numerical input, and it is a simple and effective way to represent categorical variables as numbers.

C. Null Value Handling

The null values can be fill up with median, mean, or most common value of each column. We can use advanced methods like KNN or regression to fill in null values without altering the meaning. Null Value Indicators: Create a new binary column indicating whether a value was null. This allows the model to learn patterns associated with missing data. The dataset has null values of 240 and 284 respectively plasma_CA19_9 and REG1A. Therefore, we have to handle this feature.

D. Feature Selection

Feature selection is about to take the most important features from and void fitting too closely to the data, and make them easier to understand. We use Chi-squared & SelectKBest to get

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the best feature. Chi-squared test evaluate the independence between each feature and the target variable for classification tasks. SelectKBest is a method in scikit-learn that helps to take the best features. It looks at each feature's relationship with the target (like if someone has a disease) and selects the top k features based on that. The most common test used is the chi-squared test for feature selection in classification tasks and the f_regression for regression tasks. After apply Chi-squared & SelectKBest we choose nine features for training our model and there the class name is diagnosis. Correlation of the nine features are shown in "Fig. 2".

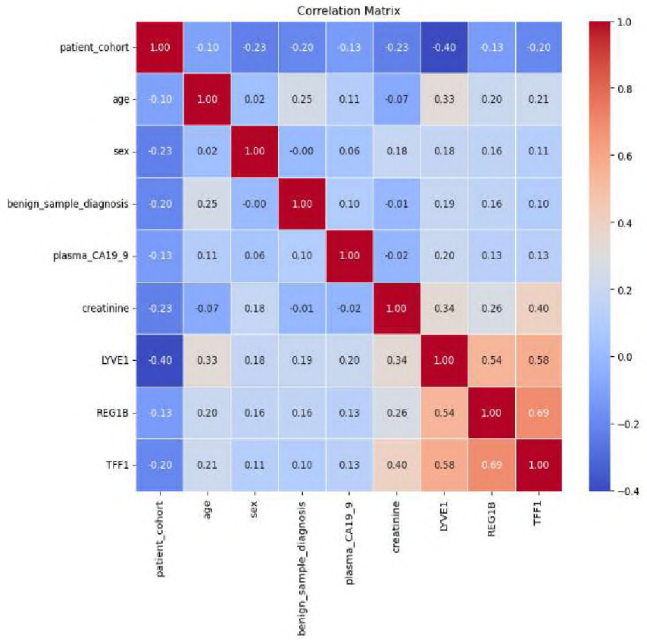


Fig. 2: Correlation of the selected features

E. Machine Learning Classifiers

Classification is important in machine learning methods to detect any disease. This section will discuss the operational principles method of RF, DT, NB, XGB, and LightGBM (LGBM) classifiers. Among these classifiers, we will find out the best classification model comparative.

Light Gradient Boosting Machine

LightGBM is a DT algorithm that employs a gradient-boosting technique to enhance predictive accuracy while optimizing computational resources [16]. In the n-occurrence training phase,stability with each symbol, which is represented by a variable. size s in space X^s, x_1, and x_n. The leaf-wise approach demonstrates reduced loss compared to the level-wise approach.

$$\tilde{V}_j(d) = \frac{1}{n}\left(\frac{\left(\sum_{x_i \epsilon A_l} g_i + \frac{1-a}{b}\sum_{x_i \epsilon B_l} g_i\right)^2}{n_l^j(d)} + \frac{\left(\sum_{x_i \epsilon r} g_i + \frac{1-a}{b}\sum_{x_i \epsilon B_r} g_i\right)^2}{n_r^j(d)}\right) \quad (1)$$

Unfavorable inclinations, represented as g_1 and g_n, are assigned for each gradient-boosting phase, characterizing either the error function or the model's response. The formation of the instance subset A involves retaining the top-a-100% instances with the greatest gradients.

3.5.2. eXtreme Gradient Boost

XGBoost, which means Extreme Gradient Boost, represents a more controlled variation of gradient boosting. This has contributed to its widespread recognition as one of the most prominent methods in the machine-learning [17]. Notably, XGBoost exhibits fast execution speed. Unlike standard gradient boosting, it adopts a distinctive technique for tree creation.

$$TheSimilarityScore = \frac{(\sum_{i=1}^{n} Residual_i)^2}{\sum_{i=0}^{n}[PreviousProbability_i * (1 - PreviousProba} \quad (2)$$

$$Gain = LEft\ leaf_{similarity} + Right\ leaf_{similarity} - Root_{similarity} \quad (3)$$

Decision Tree

In the realm of Decision Trees (DT), the nodes of the tree signify the input characteristics. The unused characteristics from preceding steps constitute the subtree. Internal nodes symbolize input components, branching signifies outputs, and leaf nodes denote classes within the Decision Tree [18]. This structure comprises multiple layers of nodes, with the root positioned at the top and the leaves situated at the bottom. Equation (4) is employed for calculating the entropy of each class, while Equation (5) computes the entropy using two characteristics.

$$H(S) = \sum_{i=1}^{c} -P_i \log_2 P_i \quad (4)$$

$$H(S, D) = \sum_{c \epsilon D} P(c).E(c) \quad (5)$$

Naïve Bayes

The Naïve Bayes algorithm, positioned within the realm of supervised machine learning, operates on Bayes' theorem with the assumption of independence among each input variable [19]. This classifier demonstrates a high degree of ease in implementation across various complex problems. Equation (6)

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serves as the means to articulate the mapping of this classifier, where $P(M|N)$ represents the posterior probability, $P(N|M)$ signifies the likelihood, $P(M)$ denotes the prior probability or initial probability, and $P(N)$ signifies the prior probability associated with the input instance.

$$P(M|N) = \frac{P(N|M) \times P(M)}{P(N)} \quad (6)$$

Extra Tree Classifier

The Extra Trees Classifier, a form of cooperative learning [20], amalgamates outcomes from several decorrelated decision trees(DT) organized in a "forest" to achieve classification proficiency. Its distinction from the random forest(RN) classifier lies primarily in the construction method of the forest's decision tree(DT) algorithm, making the two considerably comparable overall. Each decision tree within the extra trees' forest utilizes the initial training sample as its foundation. Furthermore, at each stage, every decision tree is required to select the optimal characteristic for division based on some numerical criteria from a random selection of k features provided to it. This process results in the generation of multiple decorrelated decision trees owing to the arbitrary feature sample.

$$Entropy\ (S) = \sum_{i=1}^{c} -p_i \log_2(p_i) \quad (7)$$

K-Nearest Neighbor

The K-Nearest Neighbor algorithm is a straightforward machine learning technique, renowned for its ease of implementation and heightened effectiveness with extensive training data. Primarily utilized for classification and regression within the realm of supervised techniques, it operates as a non-parametric classifier, relying on feature similarity. Predictions for the values of new data points are made based on this feature similarity. A discount is applied to the latest data points, contingent upon the proximity to matching points in the training dataset [21]. This classifier assesses distance using various formulas, with one popular formula outlined below.

$$Euclidean = \sqrt{\sum_{i=1}^{k} (x_i - y_i)^2} \quad (8)$$

IV. Result and Discussion

This section delineates the outcomes of the evolving system, with 80% data allocated for training 20% for testing in this study. Employing various machine learning methods, the approach incorporates techniques such as handling null values, label encoding, and feature selection to optimize accuracy levels.

A. Environmental setup

This research utilizes machine learning to early detection pancreatic cancer, employing a combination of high-performance computing tools, computer languages, and software. The configuration of the environment for the development of the model in this study is presented in Table 1.

TABLE I. ENVIRONMENT SETUP

Resource	Details
CPU	Intel® Core™ i5-7500 CPU @ 3.40GHz
RAM	16GB
GPU	Intel® UHD Graphics
Software	Anaconda
Language	Python

B. Confusion Matrix

A confusion matrix is a table that helps us understand how well a classification model is doing evaluating the predictive performance in classification problems. In the context of predicting pancreatic cancer, this study employed seven classifiers, each yielding diverse outcomes. Through the utilization of the confusion matrix, a comprehensive comparison was conducted for each class against the others, revealing the count of incorrectly classified samples. The performance of these models, as assessed through the confusion matrix, is visually presented in "Fig. 3'. Additionally, the effectiveness of the existing machine learning classifiers is systematically evaluated using a range of assessment parameters Equations (9 to 14).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

$$Precision = \frac{TP}{TP + FP} \quad (10)$$

$$Recall = \frac{TP}{TP + FN} \quad (11)$$

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (12)$$

$$Specificity = \frac{TN}{TN + FP} \quad (13)$$

$$False\ Positive\ Rate = \frac{FP}{FP + TN} \quad (14)$$

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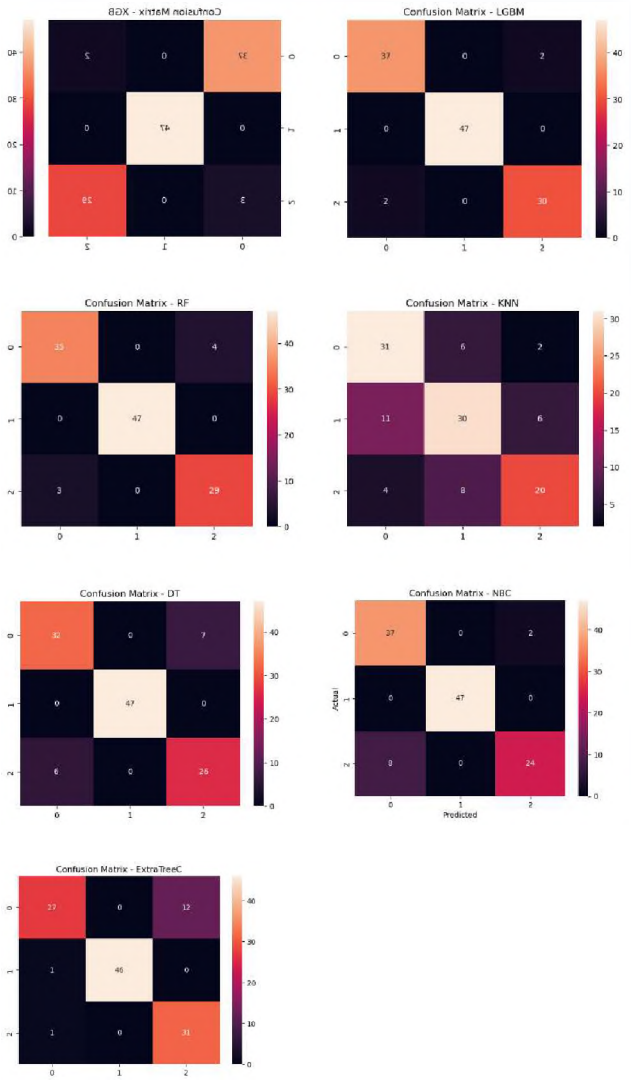


Fig.3: Confusion matrix of different ML classifiers

C. *Performance Evaluation of Machine Learning Models*

Table-2 shows the performance of different ML classifiers to predict pancreatic cancer. The Light Gradient Boosting Machine (LGBM) stood out as the best performer, scoring really high in accuracy (97.0%), precision (97.0%), recall (97.0%), and F1-score (97.0%). Accuracy tells us how often the model was correct overall, and LGBM's 97.0% score means it was pretty accurate in classifying things. Precision is about how good the model is at saying something is positive when it really is, and LGBM, XGBoost (XGB), and Random Forest (RF) all scored a strong 97.0% here, meaning they didn't often say something was positive when it wasn't.

TABLE II. ML MODEL EVALUATION

Classifiers	Precision	Recall	F1-Score
NBC	0.92	0.9	0.91
DT	0.88	0.86	0.87
KNN	0.69	0.69	0.68
RF	0.94	0.93	0.94
ExtraTree	0.89	0.87	0.87
XGB	0.96	0.96	0.96
LGBM	0.97	0.97	0.97

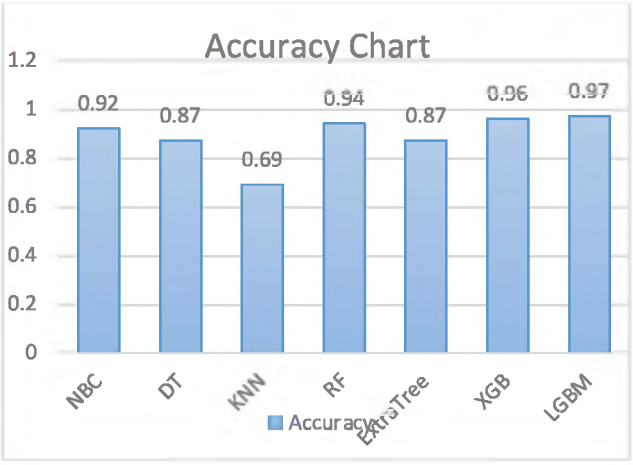


Fig. 4: Accuracy of different machine learning models

The recall score tells us how good the model is at finding all the positive cases, and again, LGBM scored a high 97.0%, showing it was excellent at capturing all the positive instances. LGBM's 97.0% in F1-score means it not only did well in finding positives but also in not wrongly saying something was positive when it wasn't. While LGBM was the top performer, it's also good to recognize that XGB and RF did well too, each scoring 96.0% in F1-score. On the other hand, DT and KNN didn't do as well, suggesting there might be room for improvement or considering other models.

In summary, LGBM is a strong choice for our work, but it's important to think about what exactly we need from our model. If we need a good balance between finding positives and avoiding mistakes, LGBM seems like the way to go.

Table 3, shows that several researchers have suggested multiple models to predict pancreatic cancer. For instance, Sadewo W. et al. [12] achieved 98% accuracy but utilized only 6 features, whereas Qiu Y. et al. [5] employed 16 features yet achieved only 70% accuracy. However, our approach utilizes the optimal feature selection approach and the maximum number of models, each achieving impressive accuracy and precision by employing 14 effective features. We obtained the highest accuracy of 91% using the LGBM classifier. So, it's very clear that our proposed method is more acceptable than the other methods that were proposed previously.

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TABLE III. COMPARATIVE PERFORMANCE ANALYSIS OF PROPOSED MODEL WITH OTHERS WORKS

Author	Methods	Features	Accuracy
Sadewo W. et al. [12]	Twin Support Vector Machine (TWSVM)	6	98%
Qiu Y. et al. [5]	DT, KNN, and SVM	16	70%
Setiawan Q. et al. [15]	DT and Logistic Regression		92.68%
Yokoyama S. et al. [14]	SVM and NNET	-	-
Proposed method	LGBM	9	97%

V. CONCLUSIONS

The pancreatic cancer affected people may not know early symptoms, often leading to detection at later stages. Despite the challenges posed by late-stage diagnosis, various treatments are available to extend the lives of affected individuals. The selection of treatments is contingent upon the extent of cancer spread throughout the body. Timely detection is crucial for enabling affected individuals to receive appropriate and early treatment, the importance of early identification. Machine learning plays a vital role in aiding the medical field by facilitating early detection. In our research paper, we worked with 590 samples. To enhance the performance of our machine learning models, we performed preprocessing on the data. The Light Gradient Boosting Machine (LGBM) stood out as the best performer, scoring really high in accuracy (97.0%), precision (97.0%), F1-score (97.0%). and recall (97.0%). As a result, our model is now well-equipped to detect PDAC earlier, showcasing a high accuracy score.

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